

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Valve spring tester, Part Number 5299714

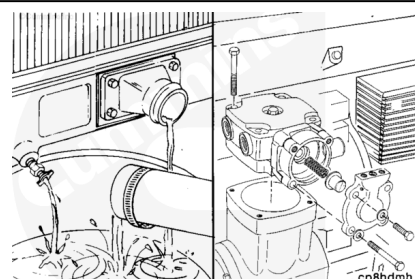
Additional Service Items

- Accrolube Lubrication Teflon™ Grease, or equivalent
- Feeler gauge

Disassemble

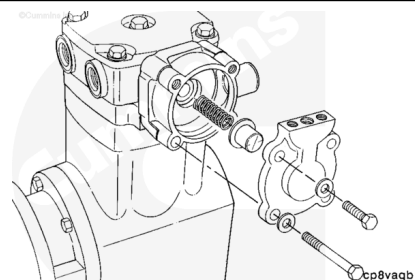
Note : If the cylinder head is removed while the air compressor is on the engine, drain the engine coolant. See equipment manufacturer service information.

Note : Since the valve plate, head, and unloader body are indexible, marking of these parts is recommended to make sure they are reassembled in the proper orientation.



⚠ WARNING ⚠

Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.



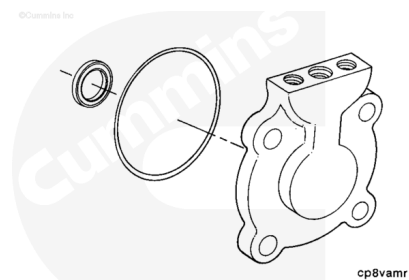
Hold the unloader valve body down. Remove the four capscrews.

Remove the unloader valve body.

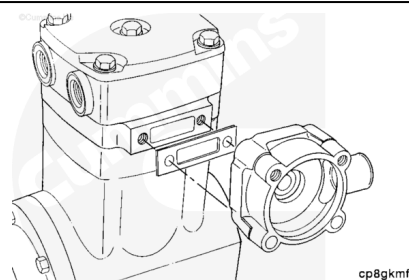
Remove the unloader valve spring.

Remove the unloader valve cap.

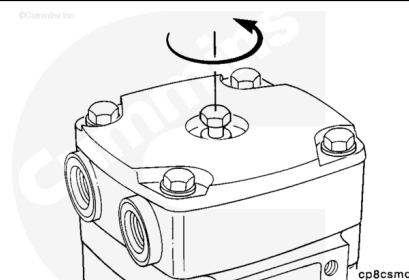
Remove the unloader body gasket and unloader valve cap rectangular ring seal.



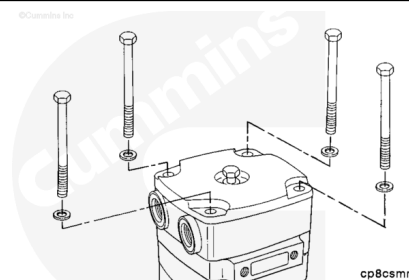
Remove the intake manifold and gasket.



Loosen, but **do not** remove, the center head capscrew.
Mark the head for orientation during assembly.

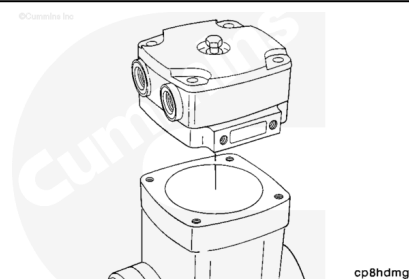


Loosen and remove the four corner head capscrews. Save the capscrews for reuse.



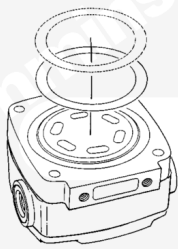
Remove the head, cover, and valve plate assembly. Place it on a clean work surface with the intake valve facing upward.

Note : If continuing with disassembly of the head, valve plate, and cover, make sure the work surface is clean. Grit pushed into the valve sealing surfaces by setting components on a dirty surface will cause a malfunction after reassembly.



Some units have a press-fit intake valve retainer. If present, carefully remove it to prevent component damage.

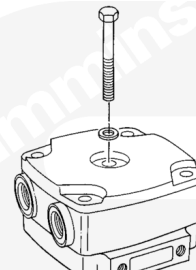
Remove the intake valve.



cp8vams

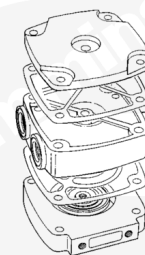
Turn the head assembly over and set it on a clean surface. Remove the center capscrew. This capscrew can be reused.

Note : The center capscrew is shorter than the four corner capscrews.



cp8hdec

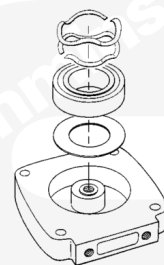
Remove the cover, cover gasket, head, and head gasket.



cp8cvmg

Remove the two wave washers, exhaust valve retainer, and exhaust valve.

Note : The wave washers must be replaced.

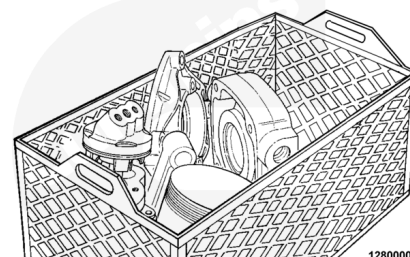


cp8wama

Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

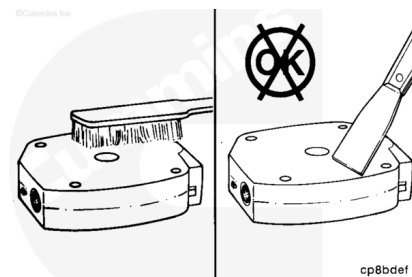


12800007

Soak the parts in a kerosene emulsion-based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier can be contacted about solution concentration, temperature and soak time.

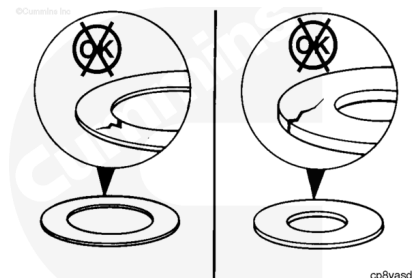
Note : Do not use a scraper to remove carbon and scale, the sealing surfaces can be damaged.

The parts can be scrubbed with a stiff nonmetallic bristle brush.



Inspect for Reuse

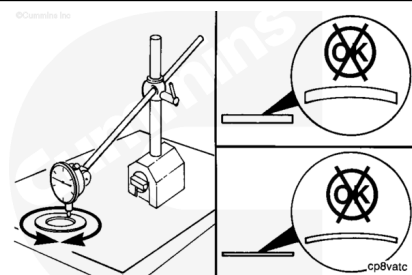
Inspect the intake and exhaust valves for cracks or other damage.



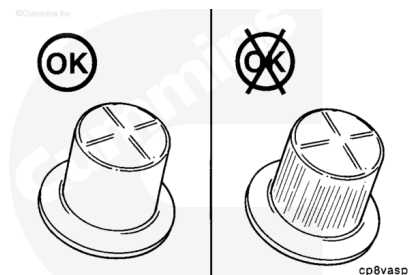
Measure the flatness of the intake and exhaust valves. Both valves **must** be flat within 0.03 mm [0.001 in].

Replace valves if cracked, damaged, or **not** flat.

Note : Holset Engineering Co., Inc., recommends new valves be installed.



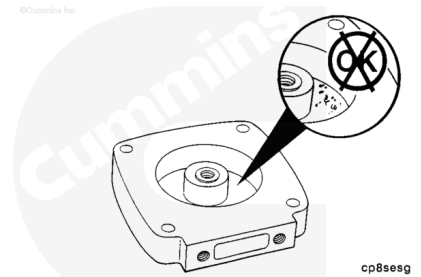
Inspect the upper part of the unloader valve cap where the rectangular V-seal operates. Check for scoring.



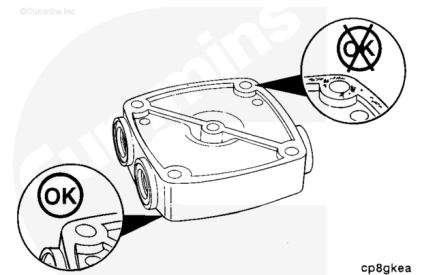
Inspect the valve seat surfaces.

Note : Inspection of the valve seats in the valve plate requires specialized equipment and is beyond the scope of field service.

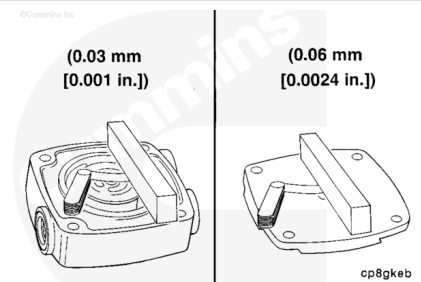
If the valve seat is visibly damaged, or **cannot** be cleaned, a new valve plate is available in a service kit. Otherwise, a QE valve plate service assembly can be used.



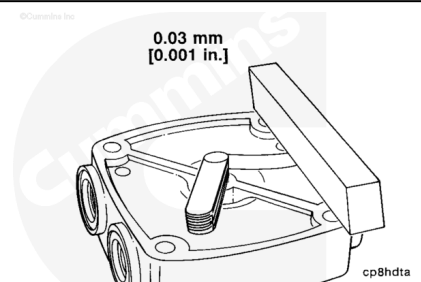
Gasket sealing surfaces **must** be clean and free of all old gasket material, carbon, rust, and other buildup. Surfaces **must** be free of scratches, gouges, burrs, and other damage.



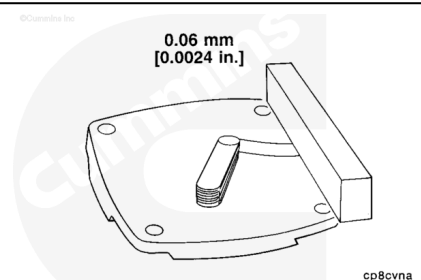
After making sure all gasket surfaces are clean and free of the above, inspect the head and cover for flatness. Use the flat plate and the feeler gauges.



Single cylinder heads **must** be flat within 0.03 mm [0.001 in] between any two adjacent capscrew holes.



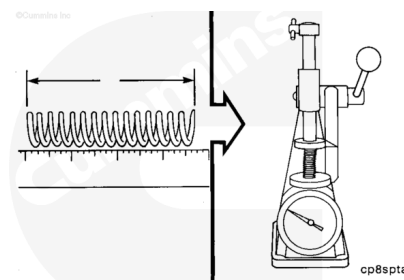
Single cylinder top cover **must** be flat within 0.06 mm [0.0024 in] between any two adjacent capscrew holes and 0.10 mm [0.004 in] total.



Use valve spring tester, Part Number 5299714, to check the unloader valve spring. Replace any spring that does **not** meet specifications.

Note : Holset Engineering Co., Inc. recommends new springs be installed.

See Compressor Spring Specifications chart shown below.



Load Required to Compress Spring to Measurement Length

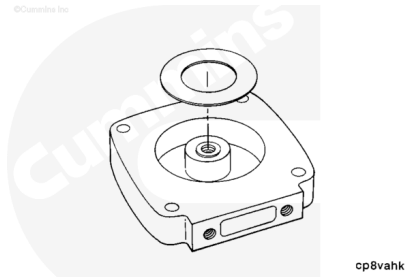
Spring Part Number	Spring Use	Approx. Free Length: mm [in]	Number of Coils	Wire Diameter : mm [in]	Measure ment Length: mm [in]	Minimu m: kg [lb]	Maximu m: kg [lb]
128080	Exhaust valve	17.02 [0.670]	3.0	2.03 [0.080]	7.11 [0.280]	3.6 [8.55]	4.7 [10.35]
190334	Intake valve	12.70 [0.500]	2.8	1.58 [0.062]	7.11 [0.280]	0.35 [0.65]	0.5 [1.10]
150631	Unloadin g valve (naturally aspirate d) center unloadin g valve - twin	41.91 [1.650]	11.5	2.03 [0.080]	24.89 [0.980]	14.5 [32.00]	17.2 [38.00]
3023101	Unloadin g valve (all turbocha rged engines)	41.91 [1.650]	10.8	1.65 [0.065]	24.89 [0.980]	5.9 [13.00]	7.7 [17.00]
3049553	E-Type unloader valve	41.91 [1.650]	11.25	1.93 [0.076]	24.89 [0.980]	10.4 [23.00]	12.2 [27.00]
800399-XW	Unloader	17.02 [0.67]	6	1.04 [0.041]	6.60 [0.260]	N/A	2.54 [5.60]
					10.03 [3.95]	1.71 [3.78]	N/A
802000-FZ	Intake and exhaust	10.16 [0.40]	4.25	0.79 [0.031]	4.57 [0.18]	N/A	0.36 [0.80]
					5.99 [0.275]	0.20 [0.45]	N/A

Load Required to Compress Spring to Measurement Length							
Spring Part Number	Spring Use	Approx. Free Length: mm [in]	Number of Coils	Wire Diameter : mm [in]	Measurement Length: mm [in]	Minimum: kg [lb]	Maximum: kg [lb]
3054489	Exhaust valve	21.49 [0.846]	4.5	2.54 [0.100]	15.21 [0.599]	3.88 [8.55]	4.74 [10.45]

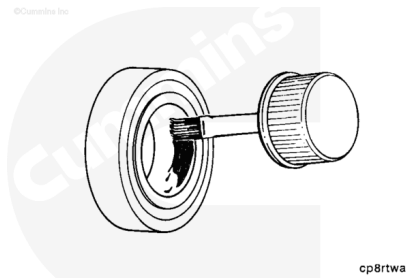
Assemble

QE, Non-European

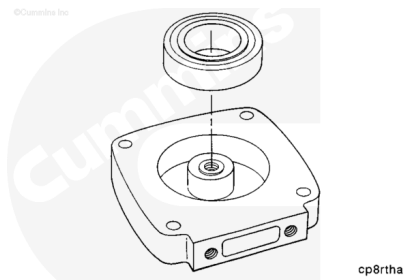
Install the exhaust valve over the post in the valve plate.



Apply a thin coating of anti-seize to the inside diameter of the exhaust valve retainer.



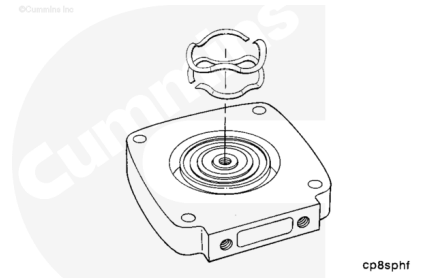
Slide the exhaust valve retainer over the valve plate. Make sure the end of the retainer with the groove faces upward.



Align the wave spring gaps 180 degrees from each other so they do **not** overlap.

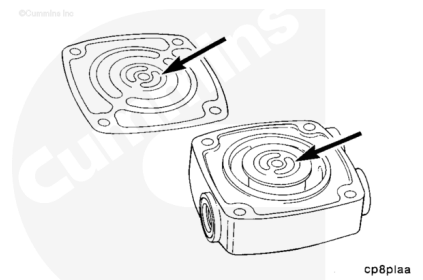


Place the wave springs in the retainer groove.



Determine the final orientation of the valve plate (air intake location) and the head (coolant ports with respect to air inlet or manifold location). Align the kidney-shaped slots in the head with the kidney shaped slots in the gasket.

If orientation marks were made before disassembly, use them.



Assemble the cover, cover gasket, head, head gasket, and valve plate.

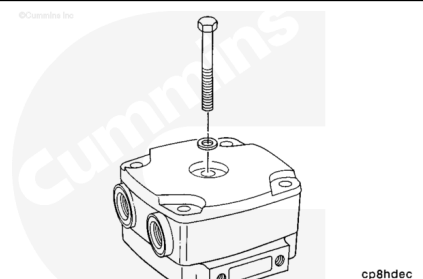


Note : Make sure corner capscrew holes are aligned.

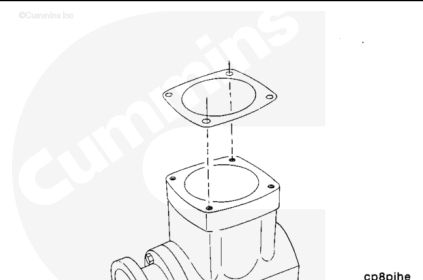
Lubricate the threads under the head.

Install the shorter capscrew with washer through the center hole.

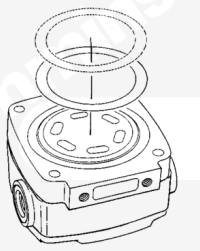
Torque Value: 14 n•m [124 in-lb]



Install the valve plate gasket.

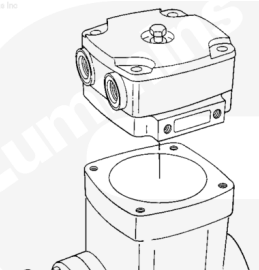


Carefully place the intake valve in the valve plate. Install the intake valve retainer.



cp8vams

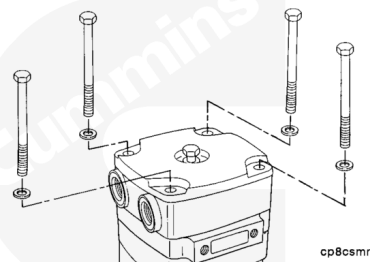
Install the valve plate assembly.



cp8hdmg

Lubricate the threads under the head and washer of the capscrews if initially installed.

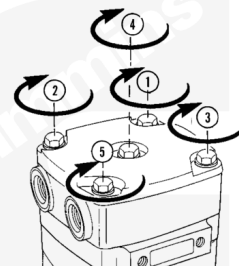
Install the four head capscrews and washers.



cp8csmr

Tighten all five capscrews.

Torque Value: 28 N·m [248 in-lb]

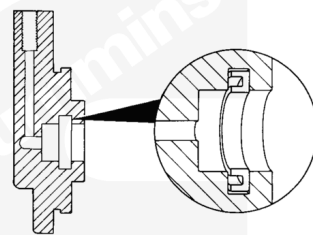


cp8cswb

Install the new rectangular V-seal into the unloader body.

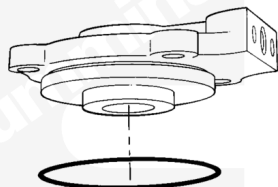
Note : The seal must be installed with the grooved side up.

Liberally lubricate the unloader valve bore above and below the rectangular ring seal with high temperature grease, Accrolube Lubrication Teflon™ Grease, or equivalent.



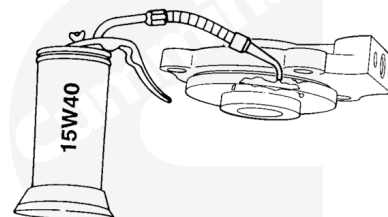
cp8seh9

Install a new o-ring seal on the unloader valve body.



cp8sehh

Use clean 15W-40 oil or Accrolube Lubrication Teflon™ Grease, or equivalent, to lubricate the seal.

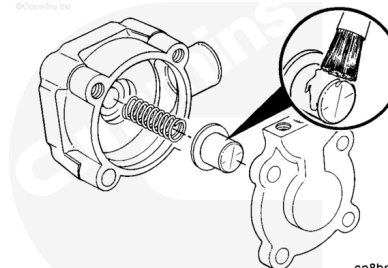


cp8sewa

Liberalily lubricate the unloader valve body bore and unloader cap with high temperature grease, Accrolube Lubrication Teflon™ Grease, or equivalent.

Install the unloader cap.

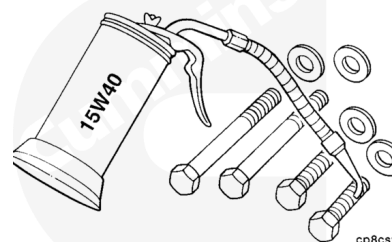
Install the unloader spring.



cp8brwb

Lubricate the unloader screw threads and underhead with clean engine oil, SAE 15W-40, before installation.

Note : The two unloader body screws must not be used to attach any brackets.

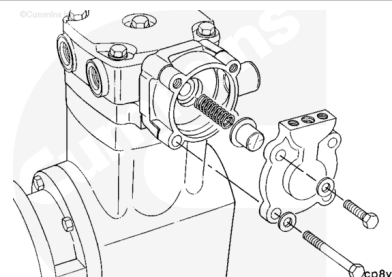


cp8cswc

Assemble the unloader components and attach the unloader assembly to the valve plate with the four capscrews and washers.

Note : The longer capscrews are used to mount the manifold to the air compressor.

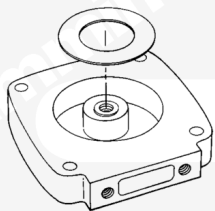
Torque Value: 27 n•m [239 in-lb]



cp8vaqb

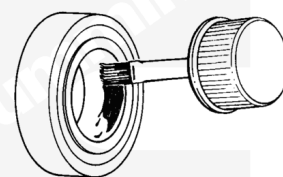
QE, European

Install the exhaust valve over the post in the valve plate.



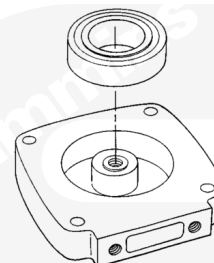
cp8vahk

Apply a thin coating of anti-seize to the inside diameter of the exhaust valve retainer.



cp8rtwa

Slide the exhaust valve retainer over the valve plate. Make sure the end of the retainer with the groove faces upward.



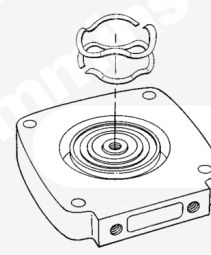
cp8rtha

Align the wave spring gaps 180 degrees from each other so they do **not** overlap.



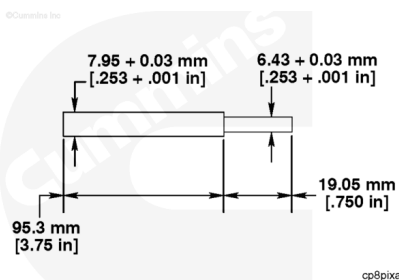
cp8sphe

Place the wave springs in the retainer groove.

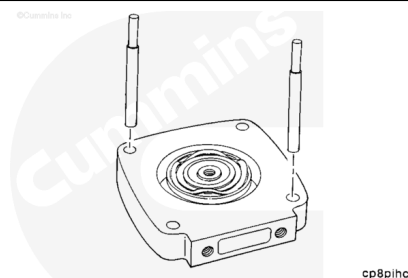


cp8spfh

Fabricate or reuse the four guide pin tools.

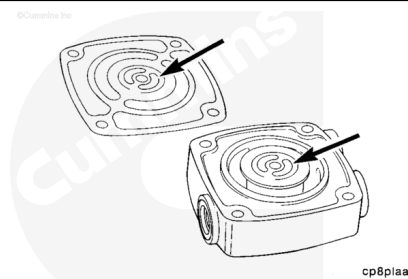


Insert the larger end of two guide pins in opposite corner holes of the valve plate.

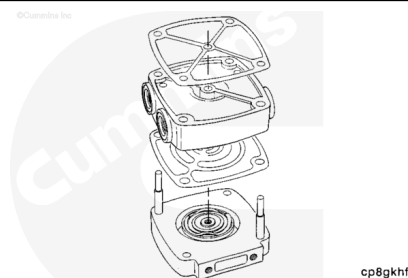


Determine the final orientation of the valve plate (air intake location) and the head (coolant ports with respect to air inlet or manifold location). Align the kidney-shaped slots in the head with the kidney shaped slots in the gasket.

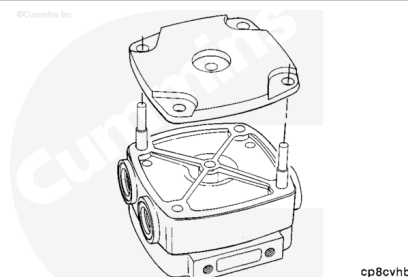
If orientation marks were made before disassembly, use them.



Install the head gasket onto the guide pins (either side up, but with correct slot orientation). Install the head onto the guide pins with the kidney-shaped slots aligned and toward the valve plate. Install the cover gasket over the guide pins.

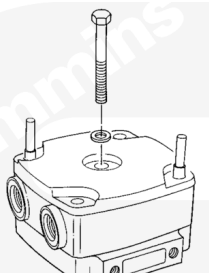


Assemble the cover.



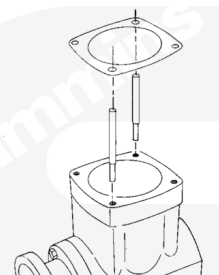
Install the shorter capscrew with washer, if initially installed, through the center hole.

Torque Value: 14 n•m [124 in-lb]



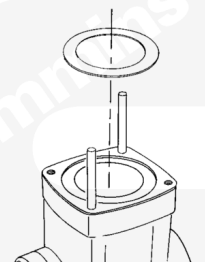
cp8cshh

Place the remaining two guide pins in the crankcase head cap screw holes (that will **not** interfere with the guide pins already in the head assembly). Install the valve plate gasket.



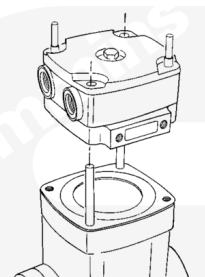
cp8pihd

Carefully place the intake valve on the crankcase, located by the valve plate gasket. Do **not** allow the valve to overlap the gasket.



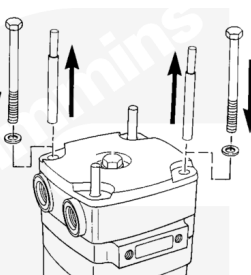
cp8vahf

Install the head assembly over the guide pins. Be careful **not** to disturb the location of the intake valve. The compressor will **not** work if the valve overlaps the gasket and is pinched.



cp8hdhb

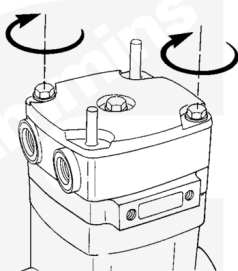
Carefully remove two of the guide pins and replace with two head capscrews and washers, if initially installed.



cp8pimb

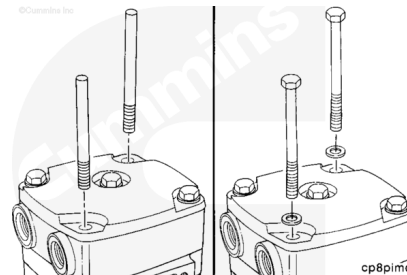
Tighten the two head capscrews.

Torque Value: 14 N·m [124 in-lb]



cp8cswa

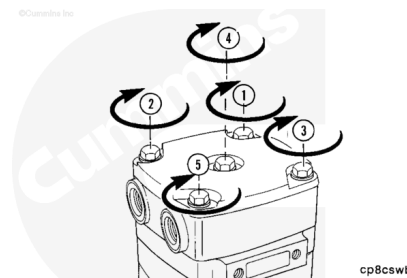
Remove the remaining two pins and replace with two head capscrews and washers if initially installed.



cp8pimc

Tighten all five capscrews.

Torque Value: 28 n•m [248 in-lb]

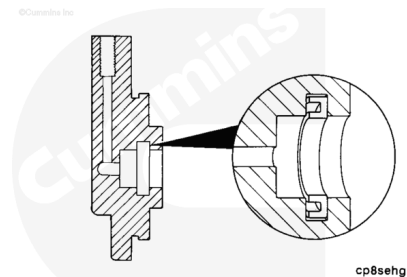


cp8cswb

Install the new rectangular V-seal into the unloader body.

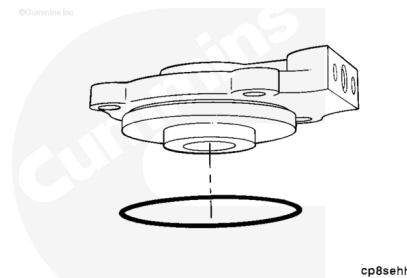
Note : The seal must be installed with the grooved side up.

Liberalily lubricate the unloader valve bore above and below the rectangular ring seal with high temperature grease, Accrolube Lubrication Teflon™ Grease, or equivalent.



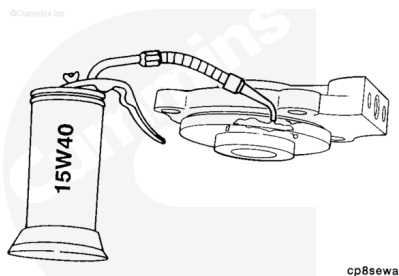
cp8sehg

Install a new o-ring seal on the unloader valve body.



cp8sehh

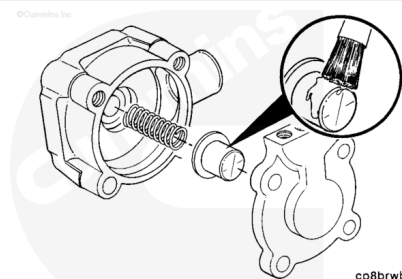
Use clean 15W-40 oil or Accrolube Lubrication Teflon™ Grease, or equivalent, to lubricate the seal.



Liberally lubricate the unloader valve body bore and unloader cap with high temperature grease Accrolube Lubrication Teflon™ Grease, or equivalent.

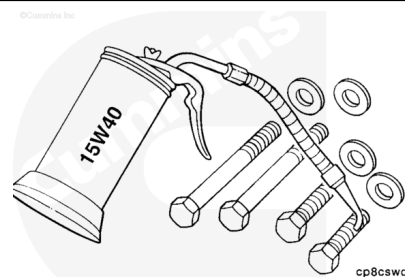
Install the unloader cap.

Install the unloader spring.



Lubricate the unloader screw threads and under the head with clean engine oil, SAE 15W-40, before installation.

Note : The two unloader body screws must not be used to attach any brackets.



Assemble the unloader components. Attach the unloader assembly to the valve plate with the four capscrews and washers.

Note : The longer capscrews are used to mount the manifold to the air compressor.

Torque Value: 27 n•m [239 in-lb]

